Deliverable 5: Final Product

Honeybee Observation Density and Urban Development Change in the San Gabriel Valley, California (2014 to 2024)

1. Introduction

Honeybees (Apis mellifera) are among the most important pollinators in the world. They support agricultural production and wild plant reproduction, and they are also common in urban environments where gardens and ornamental plants provide forage. In Southern California, urban development has continued to expand, adding more roads, rooftops, parking lots, and other paved surfaces that reduce and fragment the habitat pollinators depend on. The San Gabriel Valley is a good example of this because it has a strong urban gradient and enough public observation data to map bee activity and compare it to development patterns over time.

Research on urban pollinators shows that the relationship between urbanization and bee presence is not always negative. Wenzel et al. (2020) reviewed a large body of pollinator research across urban gradients and found that outcomes vary depending on habitat quality, green space design, and the intensity of development. Their findings support the need for local, place specific analysis rather than assuming one general trend applies everywhere. Fox et al. (2022) found that honeybees in urban environments can access diverse floral resources under the right conditions, particularly in older neighborhoods with established gardens and street trees. This is important because it means urban areas are not automatically bad for bees, which is something this project tests in the San Gabriel Valley specifically. Youngsteadt et al. (2015) linked urbanization to increased pathogen pressure on both feral and managed honeybee colonies, showing that even when bees are present in cities, the urban environment can reduce colony health over time. Campbell et al. (2023) validated iNaturalist as a source of research grade biodiversity data appropriate for spatial analysis and emphasized the importance of acknowledging sampling bias in citizen science datasets.

This project uses iNaturalist honeybee observation data and the NLCD Annual Fractional Impervious Surface product to test whether honeybee observation density in the San Gabriel Valley is associated with urban development intensity, and whether areas that became more developed between 2014 and 2024 show corresponding changes in where bees are being observed.

Research Question: In the San Gabriel Valley, does honeybee observation density change in relation to urban development intensity, and do areas that become more developed over time show declines or spatial shifts in honeybee observation density?

Hypothesis: Honeybee observation density is lower in areas with higher urban development measured by impervious surface, and areas that become more developed over time show declines or a shift in honeybee observation density compared to areas with little development change.

2. Methods

2.1 Spatial Analysis and GIS Processes

All spatial analysis was conducted in ArcGIS Pro. The workflow had four main stages: data preparation, hotspot analysis, impervious surface change analysis, and foraging range analysis.

Data Preparation. The iNaturalist honeybee observation CSV was imported using the Display XY Data tool with longitude as the X field, latitude as the Y field, and GCS WGS 1984 as the coordinate system. The resulting point layer was exported as a permanent feature class and clipped to the San Gabriel Valley boundary polygon from the SCAG Open Data Portal. A Year field was added using Calculate Field, pulling the four digit year from the observed_on date string. The full dataset has 6,370 research grade observations from 2014 to 2024. Two temporal subsets were created: an early period from 2014 to 2018 with 1,642 observations, and a late period from 2019 to 2024 with 4,728 observations.

Hotspot Analysis. Optimized Hot Spot Analysis using the Getis Ord Gi* statistic was run on the full dataset and on each temporal subset separately. This tool identifies where bee observations are clustering in statistically significant ways, meaning the concentration is greater or less than what would be expected by chance. Output grid cells are classified by confidence level at 90%, 95%, and 99%. The San Gabriel Valley boundary was used as the bounding extent for all three runs. Running the analysis on both time periods separately makes it possible to see whether hotspot locations shifted, intensified, or emerged over the study period.

Impervious Surface Change Analysis. All eleven NLCD Annual Fractional Impervious Surface rasters from 2014 to 2024 were clipped to the SGV boundary using Clip Raster. A Set Null operation was applied to the 2014 and 2024 rasters to remove NoData pixels before subtraction. Raster Calculator was then used to subtract the 2014 layer from the 2024 layer, producing a change raster where positive values mean more pavement and negative values mean less. The output was classified into three categories: decreased impervious surface, no significant change, and increased impervious surface.

Foraging Range Buffer Analysis. Western honeybees typically forage within about 3 miles of their hive. A 3 mile buffer was generated around only the statistically significant hotspot grid cells from the full period Optimized Hot Spot Analysis, using Select by Attributes to filter cells where Gi_Bin was greater than zero before running the Pairwise Buffer tool with dissolve enabled. The resulting buffer zones represent areas where source hive colonies are most likely located given where statistically significant bee observation clusters exist.

2.2 Workflow

1. Import bee observation CSV, Display XY Data, export as feature class
2. Clip bee points to SGV boundary, add Year field, create early and late period subsets
3. Clip all 11 NLCD rasters to SGV boundary, apply Set Null to remove NoData values
4. Raster Calculator: 2024 NLCD minus 2014 NLCD, classify change into 3 categories
5. Optimized Hot Spot Analysis on full dataset, early period, and late period with SGV bounding polygon

6. Select significant hotspot cells where Gi_Bin greater than 0, run 3 mile Pairwise Buffer with dissolve
7. Export all map layouts with titles, legends, scale bars, and north arrows

2.3 Datasets

iNaturalist Honeybee Observations (*Apis mellifera*). Source: iNaturalist observation export filtered to research grade quality, taxon Apis mellifera, San Gabriel Valley extent, January 1, 2014 to December 31, 2024. The dataset contains 6,370 observations with latitude, longitude, date observed, observer name, positional accuracy, and species identification. iNaturalist is a joint initiative of the California Academy of Sciences and the National Geographic Society.

NLCD Annual Fractional Impervious Surface (2014 to 2024). Source: Multi Resolution Land Characteristics Consortium, U.S. Geological Survey. Annual raster products at 30 meter resolution representing the percentage of impervious cover per pixel. Eleven annual layers were downloaded from the MRLC Viewer at mrlc.gov and clipped to the SGV boundary.

San Gabriel Valley Boundary. Source: Southern California Association of Governments Open Data Portal. The Subregions and Councils of Government dataset was used to extract the San Gabriel Valley subregion polygon, which served as the study area boundary and clip extent for all layers.

3. Results

Map 1 — Honeybee Observation Hotspots, Early Period (2014 to 2018)

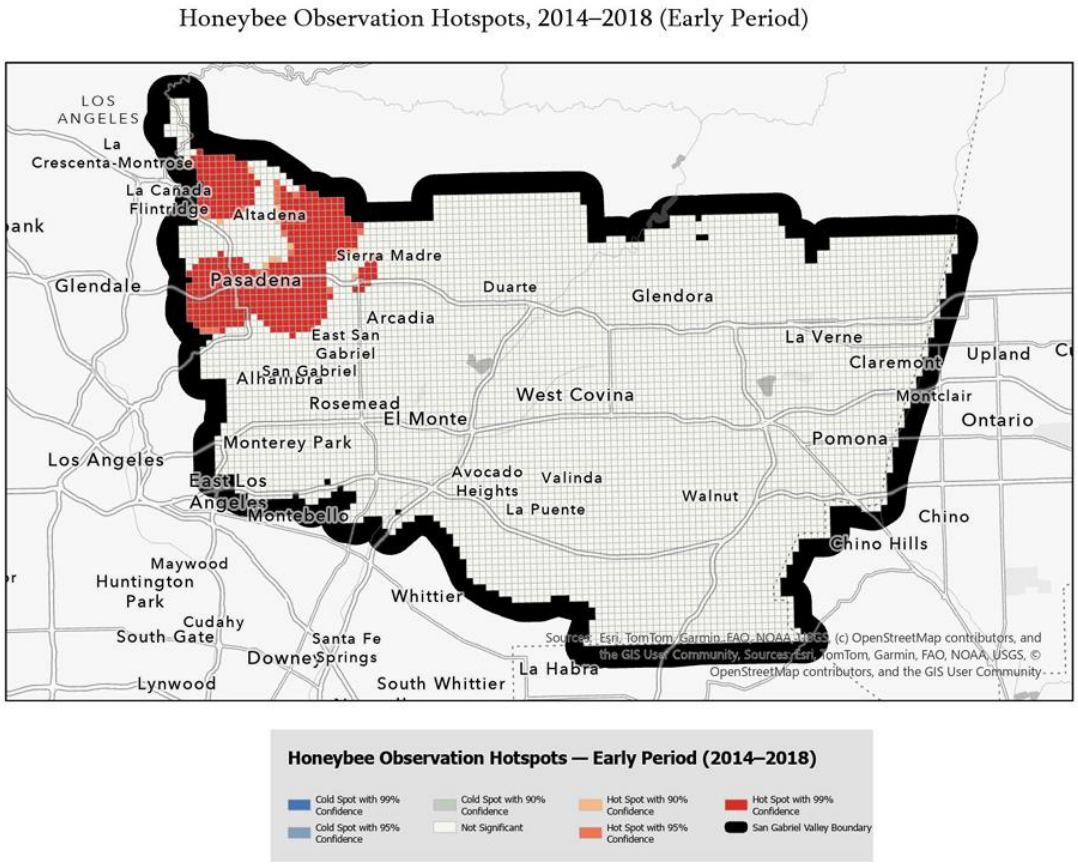


Figure 1. Optimized Hot Spot Analysis for the early period (2014 to 2018, n = 1,642 observations). Statistically significant hotspot activity at 99% confidence is concentrated in Pasadena, Altadena, and La Canada Flintridge near the San Gabriel Mountains foothills. No significant hotspot activity was detected in the central or eastern valley during this period.

Map 2 — Honeybee Observation Hotspots, Late Period (2019 to 2024)

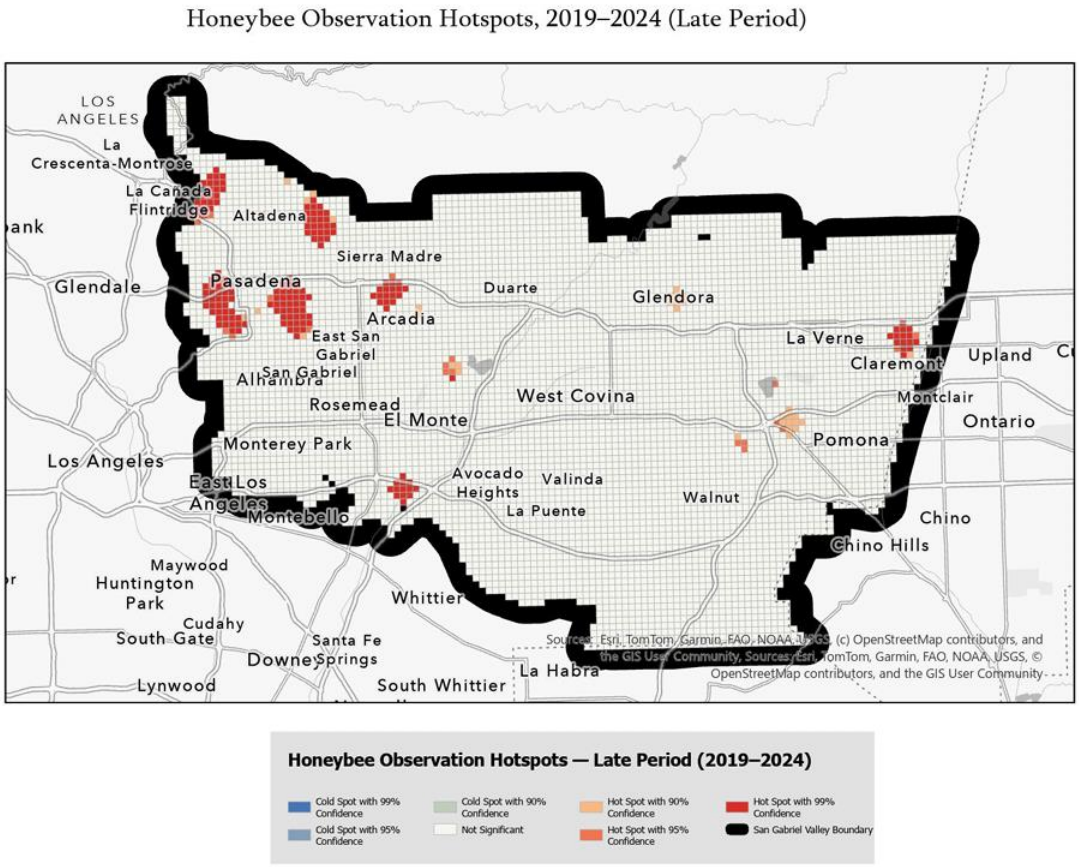


Figure 2. Optimized Hot Spot Analysis for the late period (2019 to 2024, n = 4,728 observations). The northwestern cluster in Pasadena and Altadena persists. New hotspots emerge in Arcadia, South El Monte, Claremont, and near Pomona. All new hotspot locations correspond to areas that showed no significant impervious surface change between 2014 and 2024.

Map 3 — Honeybee Observation Hotspots, Full Study Period (2014 to 2024)

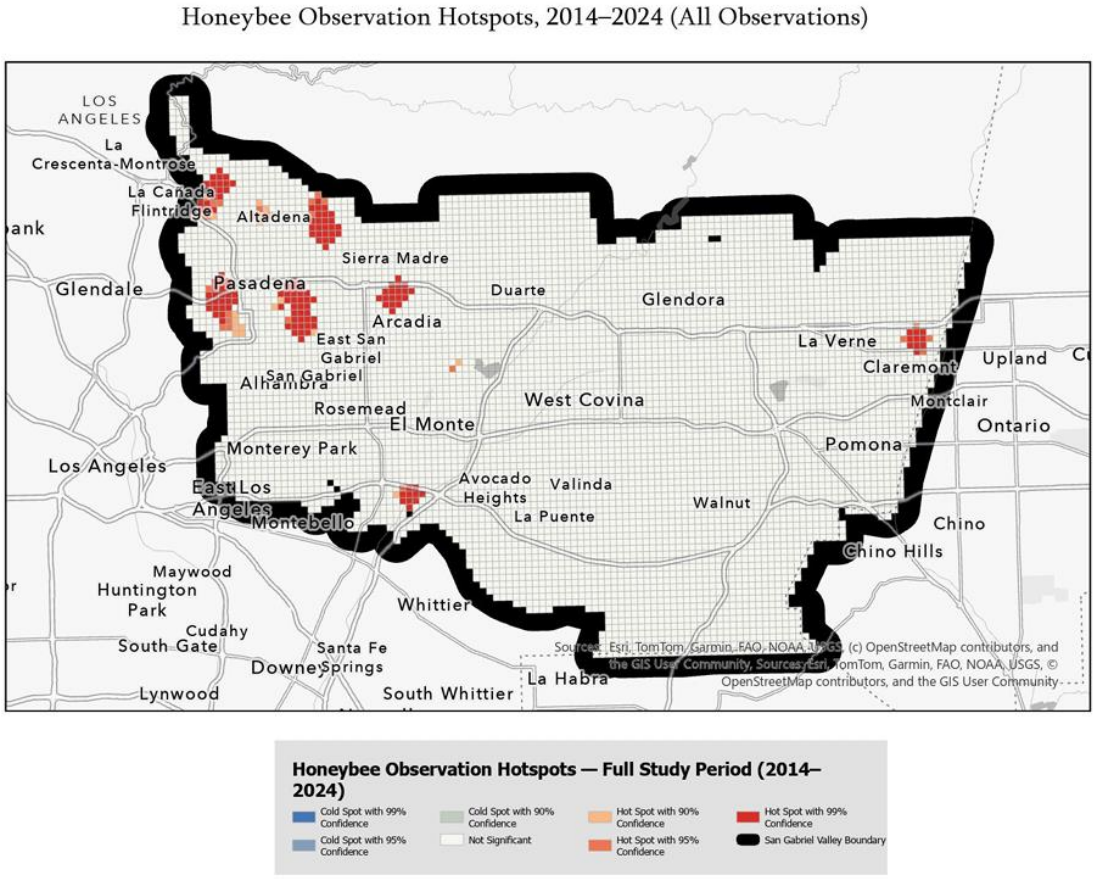


Figure 3. Optimized Hot Spot Analysis for the full study period (2014 to 2024, n = 6,370 observations). Every statistically significant cluster at 99% confidence corresponds to a known green institutional space: the Arroyo Seco corridor and Eaton Canyon Natural Area in Pasadena and Altadena, the Los Angeles County Arboretum and Botanic Garden in Arcadia, and the California Botanic Garden in Claremont. The eastern valley including Walnut, Diamond Bar, and Chino Hills shows no significant hotspot activity in either time period.

Map 4 — Impervious Surface Change (2014 to 2024)

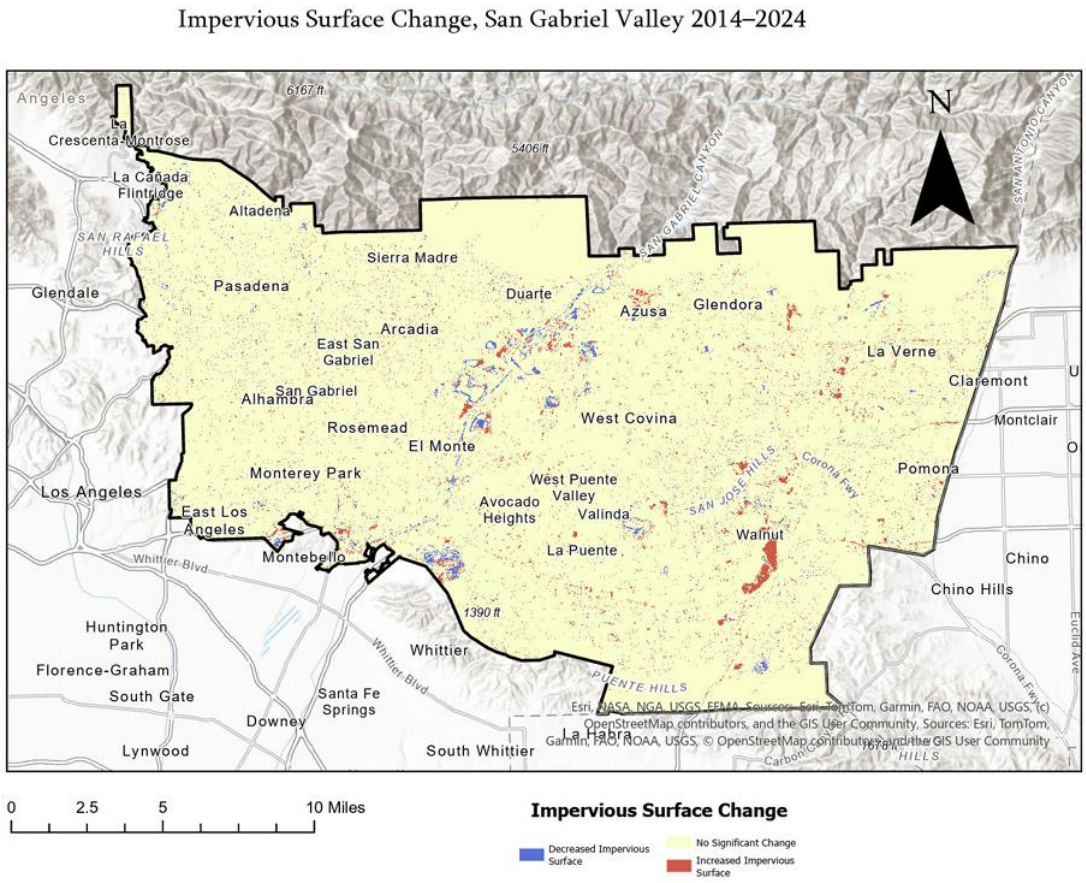


Figure 4. Change in fractional impervious surface across the San Gabriel Valley from 2014 to 2024. The majority of the valley shows no significant change, consistent with a landscape that was already heavily developed by 2014. The largest increases in impervious surface are concentrated in the Walnut and Diamond Bar area near the 57 and 60 freeway interchange, in La Verne and San Dimas, and in scattered parcels around Glendora and Azusa. Decreases in impervious surface appear in the Arcadia, El Monte, and Montebello areas.

Map 5 — Foraging Range Analysis (3 Mile Buffer)

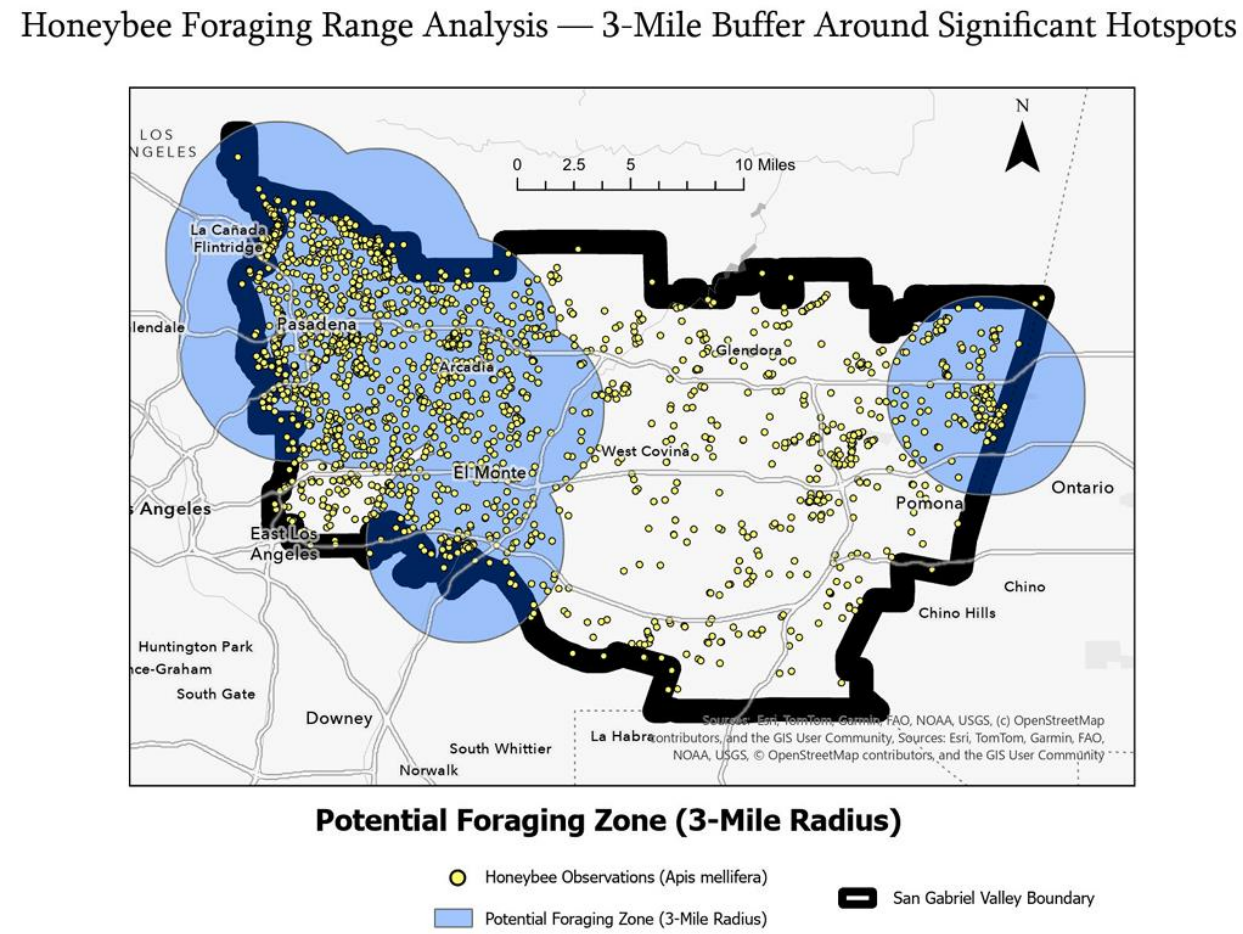


Figure 5. Three mile foraging range buffers drawn around statistically significant hotspot clusters from the full period analysis, overlaid on all bee observation points. The western buffer zone covers Pasadena, Altadena, Arcadia, and South El Monte as one large connected foraging area. A separate eastern buffer covers the Claremont and La Verne area. The central and eastern valley including West Covina, Walnut, and Diamond Bar falls entirely outside both buffer zones, meaning bee observations in those areas are not associated with any established statistically significant colony. These scattered points likely represent foragers traveling from the western hive zone or isolated individuals with no local anchor.

4. Discussion

The results support the hypothesis that honeybee observation density is lower in areas with higher urban development, but the pattern is more specific than I originally expected. It is not simply that paved surfaces push bees out. What the data actually shows is that bee activity in the San Gabriel Valley is anchored to specific large green institutional spaces, and the presence or absence of those anchors is what determines where statistically significant clusters appear.

Every hotspot at 99% confidence in the full period analysis lines up with a known botanical garden or natural area. The Pasadena and Altadena cluster corresponds to the Arroyo Seco corridor and Eaton Canyon Natural Area. The Arcadia cluster sits on the Los Angeles County Arboretum and Botanic Garden. The Claremont cluster aligns with the California Botanic Garden. These are not just low impervious areas in general. They are managed, resource rich green spaces that provide consistent floral diversity year round. The eastern valley has no equivalent institutions, and it has no hotspots.

The temporal comparison between the early and late periods adds another layer to this. In the early period almost all significant activity was in the northwestern corner near the foothills. By the late period that core cluster was still there and new ones emerged in Arcadia and Claremont. What is important is that those new hotspots did not appear in places that gained development. They appeared in places that stayed stable, where established vegetation had time to support consistent foraging. That is consistent with the foraging range buffer results, which show that the central and eastern valley is outside any established colony zone even though scattered bee observations do exist there.

The foraging range analysis also adds ecological grounding to the hotspot findings. Because honeybees typically forage within 3 miles of their hive, the buffer zones show where source colonies are most likely located. The bee observations outside the buffer zones in the central and eastern valley are likely foragers that traveled in from the western hive zones, or individuals from isolated managed colonies with no strong local anchor. The fact that the eastern development corridor falls entirely outside both buffer zones suggests that the habitat there is not supporting established wild or feral colonies in any sustained way.

One limitation to acknowledge is the sampling bias in iNaturalist data. Observations cluster where people spend time and where more users are active, which tends to favor the western valley and older residential neighborhoods. The hotspot analysis normalizes for overall observation density, so it is not just identifying where users are concentrated, but it does not eliminate the bias entirely. Future research could apply observer effort normalization or compare iNaturalist patterns to structured survey data to test whether the hotspot locations reflect genuine ecological patterns or partially reflect user distribution.

Overall this project shows that protecting large managed green spaces in the San Gabriel Valley matters for maintaining bee habitat. The botanical gardens and natural corridors identified in this analysis are not just recreation assets. They are functioning ecological refuges in an otherwise heavily developed landscape. As development pressure continues in the eastern valley, the absence of equivalent anchor green spaces there suggests that bee habitat in that part of the SGV is already limited and unlikely to support established colonies without intentional conservation or restoration efforts.